



ECE 501: Digital Image Processing

Week 4 Progress Report

Group 6

Vehicle Orientation Estimation Using Conventional Image Processing

Submitted to faculty: Prof. Mehul Raval

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GitHub Repository (DRASHTI-HaOBB):

<https://github.com/ERYAGNIK003/DRASHTI-Dataset>

Dataset Source:

DRASHTI-HaOBB – Heading-angle Oriented Bounding Box Dataset

<https://zenodo.org/records/18278989>

1. Work Done This Week

In Week 4, we focused on analysing the results obtained from the three conventional image-processing methods implemented in Week 3. The main objective was to compare the performance of PCA-based, Hough Transform-based, and contour-based orientation estimation on the same set of 100 vehicle images.

Instead of repeating the dataset extraction and orientation-estimation process, we used the Week-3 CSV file containing the ground-truth orientations, predicted orientations and angular errors for the three methods. This allowed us to concentrate on interpreting the existing experimental results.

First, an overall method-comparison table was generated. It included the number of successful and failed predictions, the success rate, the mean angular error, the median angular error, the standard deviation, and the maximum angular error for each method.

Next, four graphs were generated to visualise the results. These included a mean angular error comparison, an angular-error boxplot, a combined ground-truth versus estimated orientation plot with separate panels for the three methods, and a class-wise mean angular error comparison.

A second table was created to compare the performance of PCA, Hough Transform and contour-based methods across the vehicle classes present in the selected sample. This helped identify classes for which the methods produced different levels of orientation-estimation error.

The analysis showed that PCA had the lowest mean angular error among the three methods. The contour-based method also produced a similar mean error, while Hough Transform had a higher mean error and failed to produce valid predictions for two samples.

The complete Week-4 workflow was:

Week-3 CSV → Data Loading and Validation → Overall Method Comparison → Error Distribution Analysis → Ground-truth versus Estimated Orientation → Class-wise Analysis → Graphs and Tables → Saving Results to Google Drive.

2. Methodology

The Week-4 methodology focused on the quantitative and visual analysis of the results obtained in Week 3.

2.1 Loading and Validating the Results

The saved Week-3 CSV file was loaded into Python using the Pandas library. It contained the vehicle identifiers, vehicle classes, difficulty values, ground-truth headings, converted ground-truth orientations, estimated orientations and angular errors for PCA, Hough Transform and contour-based estimation.

The file contained 100 vehicle samples, with all selected vehicles having a difficulty value of 0. The data was checked to ensure that the required columns were available before carrying out the analysis.

2.2 Overall Method Comparison

The three methods were compared using the following performance measures:

- **Successful predictions:** Number of vehicles for which a valid orientation estimate was produced.
- **Failed predictions:** Number of vehicles for which a valid orientation estimate was not produced.

- **Success rate:** Percentage of selected vehicles for which the method produced a valid estimate.
- **Mean angular error:** Average angular error across all valid predictions.
- **Median angular error:** Middle value of the angular errors, representing the typical error more robustly when some values are very large.
- **Standard deviation:** Measure of the variation in angular errors.
- **Maximum angular error:** Largest observed angular error for a method.

The angular error for each prediction was calculated as:

$$Angular\ Error = | \theta_{predicted} - \theta_{ground\ truth} |$$

Both the estimated orientation and ground-truth orientation were represented in the required 0°–90° range.

The mean angular error was calculated by averaging the valid angular-error values for each method. A lower mean angular error indicates closer average agreement with the ground truth.

2.3 Error Distribution Analysis

A boxplot was generated to compare the distributions of angular errors for all three methods. The boxplot helps compare the median, spread and potential outliers of each method.

This analysis is important because two methods can have similar mean errors but different distributions. A method with a large spread of errors may perform well on many vehicles while producing very large errors on a smaller number of samples.

2.4 Ground-truth versus Estimated Orientation

A combined three-panel graph was generated, with separate panels for PCA, Hough Transform and contour-based estimation.

In each panel, the ground-truth orientation is plotted on the x-axis and the estimated orientation on the y-axis. A diagonal reference line represents ideal estimation, where the estimated orientation is equal to the ground-truth orientation.

Points closer to this line indicate closer agreement between the predictions and the ground truth. The graphs provide a visual complement to the numerical angular-error measurements.

2.5 Class-wise Performance Analysis

The 100-image results were grouped by vehicle class. The mean angular error was calculated for each class of vehicle.

This comparison was used to examine whether the methods behaved differently for different vehicle types. The number of available samples in each class was also considered when interpreting the results, as some classes had very few examples.

3. Results and Observations

The Week-4 analysis was carried out on the same 100 vehicle samples selected and processed in Week 3. The sample represented 11 vehicle classes and was not equally distributed across the classes.

SUV was the most frequent class with 34 samples, followed by MotorCycle with 30 samples. AutoRicksaw had 8 samples, Sedan had 7, Truck had 5 and PickUp had 4. Auto3WCargo, Container and Trailer each had 3 samples, Bus had 2 samples and Tanker had 1 sample. Mixer, Tipper and Van were not present in this particular sample.

3.1 Overall Method Comparison

The overall performance obtained from the Week-4 analysis is presented in Table 1.

Method	Successful Predictions	Failed Predictions	Success Rate	Mean Angular Error	Median Angular Error	Standard Deviation	Maximum Error
PCA	100	0	100%	10.74°	4.85°	16.11°	81.71°
Hough Transform	98	2	98%	12.46°	4.18°	20.44°	82.39°

Contour-based	100	0	100%	10.97°	5.72°	16.44°	89.00°
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Table 1. Overall comparison of the three orientation-estimation methods.

PCA achieved the lowest mean angular error (10.74°) with a 100% success rate. Contour-based estimation showed a similar mean error (10.97°), while Hough Transform had a higher mean error (12.46°) and two failed predictions. Despite its lowest median error, Hough Transform showed the greatest variability. The large maximum errors indicate that all three methods had some challenging cases.

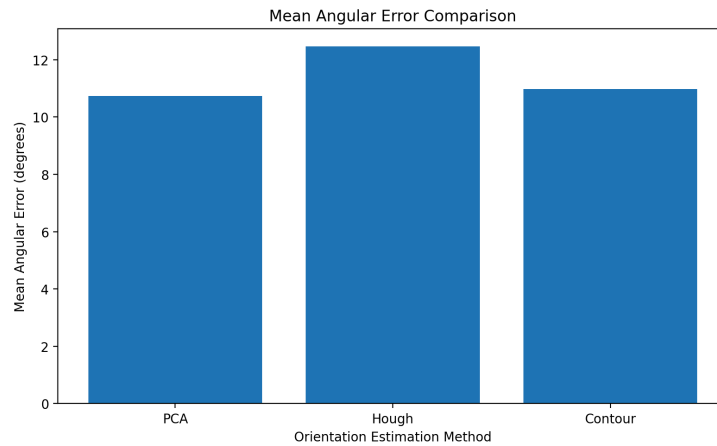


Figure 1. Comparison of the mean angular errors of PCA, Hough Transform and contour-based orientation estimation.

3.2 Error Distribution

Figure 2 compares the distribution of angular errors across the methods. Hough Transform had the lowest median error (4.18°), followed by PCA (4.85°) and contour-based estimation (5.72°). However, its higher mean error and standard deviation indicate greater variability. The boxplot also helps identify outliers and differences in error spread.

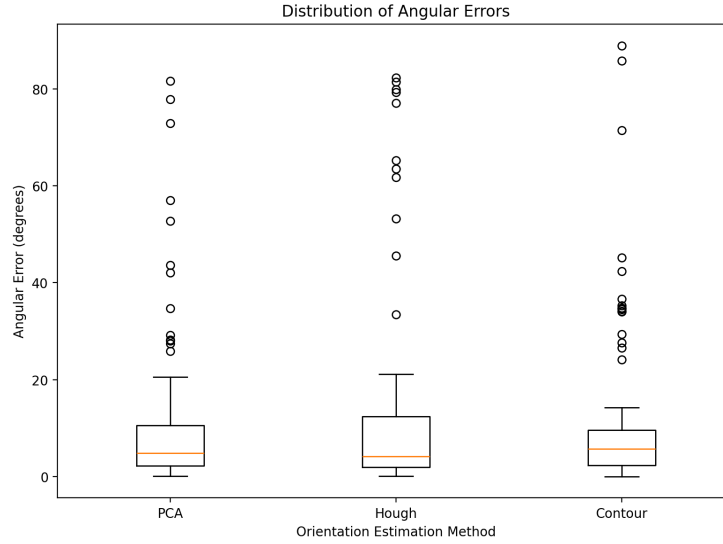


Figure 2. Distribution, spread and potential outliers of the angular errors produced by the three methods.

3.3 Ground-truth versus Estimated Orientation

The three-panel ground-truth-versus-estimated orientation graph was generated to compare the predictions of PCA, Hough Transform, and the contour-based method.

The diagonal reference line represents perfect agreement between the ground-truth and estimated orientations. Deviations from this line represent differences between the predicted and actual orientations.

This graph complements the overall error table by showing the individual prediction behaviour across the 0° – 90° orientation range. The numerical results confirm that the three methods produced different errors on individual vehicles, with the maximum angular errors reaching 81.71° for PCA, 82.39° for Hough Transform and 89.00° for contour-based estimation.

These large-error cases show that the initial methods do not reliably estimate every vehicle's orientation and that further investigation of failure cases is necessary.

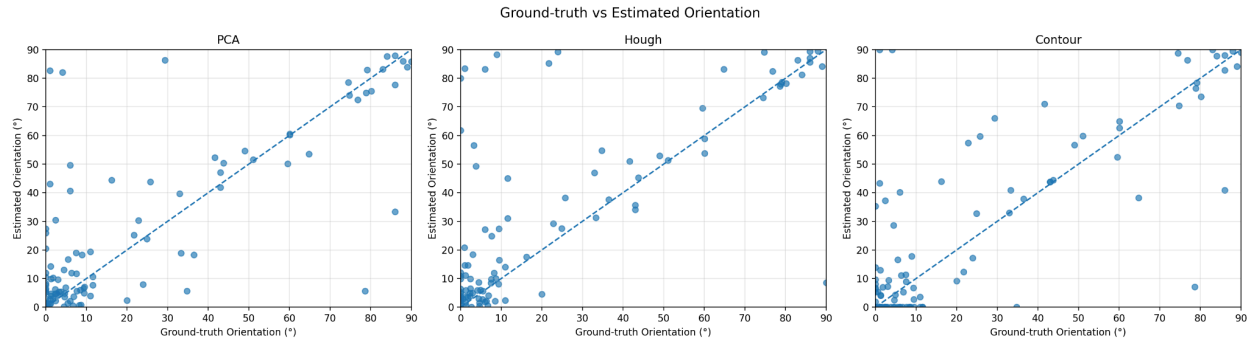


Figure 3. Ground-truth versus estimated orientation for PCA, Hough Transform and contour-based estimation.

3.4 Class-wise Performance

The class-wise comparison was used to identify differences in mean angular error across the vehicle classes represented in the sample.

The results are presented in the table below.

Vehicle Class	Samples	PCA Mean Error	Hough Mean Error	Contour Mean Error
Auto3WCargo	3	11.34°	38.58°	13.41°
AutoRicksaw	8	7.49°	13.97°	8.29°
Bus	2	27.32°	0.74°	24.18°
Container	3	40.12°	49.13°	27.49°
MotorCycle	30	7.28°	4.54°	10.15°
PickUp	4	8.73°	20.72°	3.86°
SUV	34	7.64°	12.97°	8.34°
Sedan	7	23.55°	7.03°	18.10°
Tanker	1	0.19°	3.27°	7.00°

Trailer	3	28.97°	32.87°	30.21°
Truck	5	8.05°	17.50°	6.39°

Table 2. Class-wise comparison of mean angular errors for the three methods.

For MotorCycle and SUV, PCA showed relatively low mean angular errors of 7.28° and 7.64°, respectively. The contour-based method produced mean errors of 10.15° for MotorCycle and 8.34° for SUV. Hough Transform performed well on the MotorCycle samples, with a mean error of 4.54°, but its mean error for SUV was 12.97°.

The methods behaved differently for Sedan. Hough Transform produced a mean error of 7.03°, compared with 23.55° for PCA and 18.10° for the contour-based method.

For Container, the mean errors were comparatively high for all three methods: 40.12° for PCA, 49.13° for Hough Transform and 27.49° for contour-based estimation. The contour-based method produced the lowest mean error among the three for this class.

For PickUp, contour-based estimation showed the lowest mean error at 3.86°, while Hough Transform had a mean error of 20.72°. For Truck, the contour-based method produced a mean error of 6.39°, followed by PCA at 8.05° and Hough Transform at 17.50°.

These differences indicate that the relative performance of the methods varies across vehicle classes. However, some classes have very few samples. For example, Tanker has only one sample, and Bus has only two samples. Therefore, these results should be treated as preliminary observations rather than final conclusions about the corresponding vehicle classes.

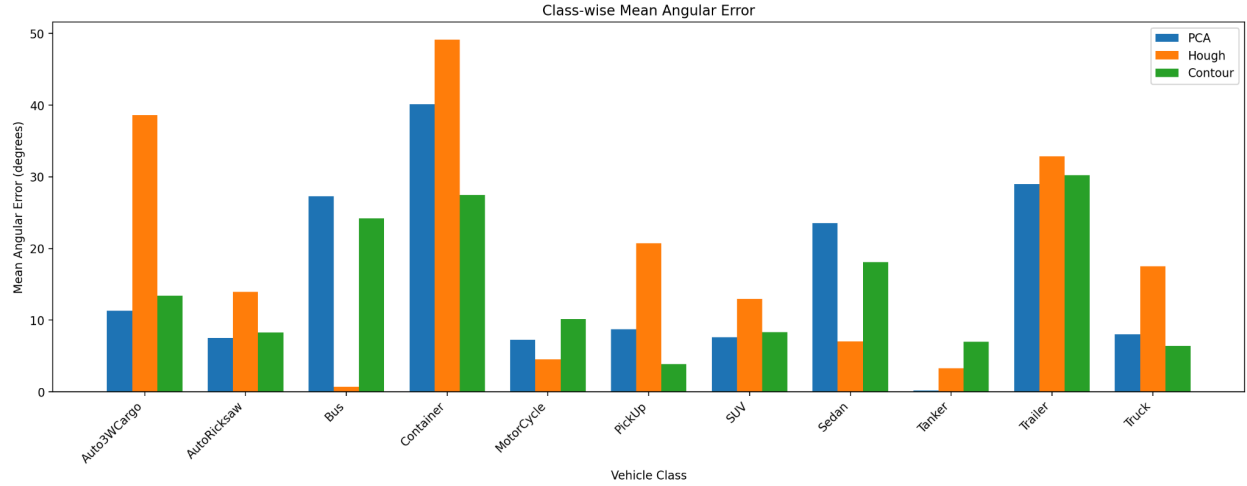


Figure 4. Comparison of mean angular error across vehicle classes represented in the 100-image sample.

3.5 Main Observations from Week 4

Based on the overall and class-wise analysis, the main observations are:

1. PCA produced the lowest mean angular error of 10.74° and generated valid estimates for all 100 samples.
2. Contour-based estimation yielded a similar mean angular error of 10.97° and produced valid estimates for all 100 samples.
3. Hough Transform produced a higher mean angular error of 12.46° and failed to generate valid estimates for two samples.
4. Hough Transform had the lowest median error of 4.18° , showing that its typical error was lower than its mean error would suggest. However, its standard deviation and mean error were the highest among the three methods.
5. Class-wise results showed different performance patterns. No method had the lowest mean angular error across all vehicle classes.
6. Some classes, particularly Container and Trailer in this sample, produced relatively high errors across the methods.
7. The uneven class distribution and the small number of examples for some classes limit the conclusions that can be drawn from the current experiment.

8. The large maximum errors indicate that the basic orientation-estimation methods still require improvement.

4. Challenges Faced

The main challenge during Week 4 was interpreting the different performance measures together. The method with the lowest mean error did not also have the lowest median error, which showed the importance of analysing the complete error distribution rather than relying on a single metric.

Another challenge was understanding the variation in performance across vehicle classes. The three methods produced different results for MotorCycle, SUV, Sedan, Container and other vehicle types. This suggests that the geometry visible in a vehicle crop and the quality of its edges and contours may influence the orientation estimate.

The sample also had an uneven class distribution. Some classes had many samples, while others had only one or two. This makes it difficult to draw reliable class-specific conclusions from the current results.

Finally, the large maximum errors indicate that some vehicle crops were not estimated accurately by the existing methods. These cases need to be inspected before deciding how to improve the preprocessing or orientation-estimation algorithms.

5. Current Status

The initial analysis of the Week-3 orientation-estimation results has now been completed.

We have:

- Loaded and validated the CSV containing the results of the initial 100-image vehicle orientation-estimation experiment.
- Generated the overall method-comparison table.
- Calculated success rates, mean errors, median errors, standard deviations and maximum errors.
- Generated the mean angular error comparison graph.

- Generated the angular-error boxplot.
- Generated the combined ground-truth versus estimated orientation graph.
- Generated the class-wise mean angular error graph.
- Created the class-wise method-comparison table.
- Saved the four graphs and two tables in the Week4_Results folder in Google Drive.

The analysis shows that PCA produced the lowest mean angular error in the current sample, while contour-based estimation produced a similar mean error. Hough Transform had a higher mean error and two failed predictions. The results also show that method performance varies across vehicle classes.

The current work provides a baseline for deciding which aspects of the orientation-estimation pipeline need improvement. It does not yet constitute a final evaluation of the complete DRASHTI-HaOBB dataset.

6. Work to Be Done Next Week

In the next stage, we plan to:

- Inspect vehicle images with large angular errors to identify possible causes of incorrect estimates.
- Improve the preprocessing and vehicle contour/edge extraction where necessary.
- Investigate suitable parameter changes for PCA, Hough Transform and contour-based estimation.
- Test the improved methods on the existing sample and compare the results with the current baseline.
- Begin the error analysis with respect to vehicle size, image quality, and ground-truth orientation.
- Consider increasing the sample size or selecting a more balanced sample for class-wise evaluation.
- Continue documenting the methodology, results and observations for the final project report.

7. References

- [1] Y. Bhavsar, M. Zaveri, M. Raval, K. R. Patel, and S. B. Zaveri, "Descriptor: Drone Nadir-view Annotated Images of Vehicles Detection Dataset for India with Heading-angle Oriented Bounding Box (DRASHTI-HaOBB)," *IEEE Data Descriptions*, 2026. Available: <https://ieeexplore.ieee.org/document/11421776>.
- [2] Y. Bhavsar, M. Zaveri, M. Raval, S. Zaveri, and Ahmedabad University, "DRASHTI-HaOBB: Drone nadir-view Annotated Images of Vehicles Dataset for India - Heading-angle Oriented Bounding Box," *Zenodo*, 2026. Available: <https://zenodo.org/records/18278989>.